

line consumption expenditure class, at the borderline separating the poor from non-poor. The Expert Committee's proposed price indices are based on the household level unit value obtained from 61st round NSS household consumption expenditure survey for different food and non-food items for both rural and urban areas. As Tendulkar Committee adopted new reference basket and new price indices, it is not comparable with the official head count ratio released for the earlier years. Table 21.2 explains the poverty line and percentage of population below poverty line. The percentage of population below poverty line declined to 21.9 per cent in 2011-12 from 37.2 per cent in 2004-05 in India. In rural India the proportion is comparatively higher as compared to urban areas. A 12 percentage point difference in poverty head count ratio is found between rural and urban areas in the year 2011-12. Between the year 2004-05 to 2011-12 a 16 percentage point reduction is visible in rural areas whereas the same for urban area is 12 percentage point.

Table 21.2: Poverty Line (Rs. Monthly Per Capita)

	Poverty Line (Rs)		Poverty Head Count Ratio (per cent)		
Year	Rural	Urban	Rural	Urban	Total
2004-05	446.7	578.8	41.8	25.7	37.2
2009-10	672.8	859.6	33.8	20.9	29.8
2011-12	816.0	1000.0	25.7	13.7	21.9

Source: Calculated from different NSS consumption expenditure data.

a) Income Poverty Indicators

Over the time, India has shown considerable reduction in poverty which is well documented. During the period between 1973-74 and 2011-12, the incidence of poverty declined continuously from 37.2 per cent to 21.9 percent or the absolute number of poor decreased from 321.3 million in 1973-74 to 269.8million in 2011-12. In rural areas the poverty level reduced from 56.4 percent to 25.7 percent and in urban areas the same has reduced from 49 percent to 13.7 percent.

However, one thing should be kept in mind that the pace of reduction in poverty varies considerably during this period with a large decline in 2011-12 and a very small decline in 1987-88. The number of people below the poverty line increased by 7.6 million during the 1973-74 to 1977-78; decreased by 21.8 million during the 1983 to 1987-88 and by 6.4 million during 1987-88 to 2004-05 (see Table 21.3). A decline of 137 million was registered between 2004-05 and 2011-12. The annual average decline (per cent) from 1993-94 to 2004-05 is 0.74 per cent, whereas the decline between 2004-05 to 2011-12 is 2.18 percent.

Table 21.3: HCR Poverty in India, 1973-74 — 2011-12

Year	Head Count Ratio (per cent)			Absolute Number of Poor below Poverty Line(in million)		
	Rural	Urban	Total	Rural	Urban	Total
1973-74	56.4	49.0	54.9	261.3	60.1	321.3
1977-78	53.1	45.2	51.3	263.3	64.6	328.9
1983	45.7	40.8	44.5	252.0	70.9	322.9
1987-88	39.1	38.2	38.9	231.9	75.2	307.1
1993-94	37.3	32.3	35.97	244.0	76.3	320.4
2004-05	41.8	25.7	37.2	326.3	80.8	407.1
2011-12	25.7	13.7	21.9	216.7	53.1	269.8

Source: Planning Commission

Considering the trend in poverty incidence, rural share of total poverty reduced from 81 per cent during 1973-74 to 80 percent (1 percentage point reduction) in 2011-12. The number of urban poor increased by 10.6 million during 1973-74 to 1987-88. Between 2004-05 to 2011-12 it decreased by 28 million in urban areas whereas in rural areas it declined by 109 million during the same period.

The poverty head count ratio by social category provides an insight of the inequality in prevalence of poverty. Highest levels of HCR among SCs and STs go with the highest depth as well severity of poverty.

The head count poverty by social category from the year 1993 to 2011-12 is provided in Table 21.4. The poverty rate among STs reduced by 17.0 percentage points (from 2004-05 to 2011-12), whereas for SCs the percentage point reduction is 21.5. The percentage point reduction for other caste population (other than SC/ST/OBC) is 10.5 percentage points. A large difference in poverty incidence is found between rural and urban areas. In 2011-12 the poverty rate of ST was 45.3 percent in rural areas whereas in urban area it was 24.1 percent (21 percentage point difference).

Poverty rates by some religion in the year 2011-12 shows that poverty rate is highest among Muslims (25.4 per cent) followed by Hindu (21.9 per cent). The rate is lowest among Jains (3.3 per cent) followed by Sikhs (5.9 per cent). It is interesting to find that the percentage point reduction in poverty head count between 2004-05 to 2011-12 shows that the reduction is highest among Muslims (43.6 per cent in 2004-05 to 25.4 per cent in 2011-12) followed by Hindu (from 37.5 per cent in 2004-05 to 21.9 per cent in 2011-12). The lowest reduction is registered among Sikhs.

Table 21.4: Poverty Rates among Social Category (1993-94 and 2011-12)

Poverty, Malnutrition
and Inclusive Growth:
Policy Implications

Area	Social Category	Percentage Population below the Tendulkar line				Percentage point poverty reduction 2004-05 to 2011-12
		1993-94	2004-05	2009-10	2011-12	
Rural	ST	65.9	62.3	47.4	45.3	16.9
	SC	62.4	53.5	42.3	31.5	22.0
	OBC	44.0	39.8	31.9	22.7	17.1
	FC		27.1	21.0	15.5	11.6
	All	50.3	41.8	33.3	25.4	16.4
Urban	ST	41.1	35.5	30.4	24.1	11.4
	SC	51.7	40.6	34.1	21.7	18.8
	OBC	28.2	30.6	24.3	15.4	15.2
	FC		16.1	12.4	8.1	8.0
	All	31.9	25.7	20.9	13.7	12.0
Total	ST	63.7	60.0	45.6	43.0	17.0
	SC	60.5	50.9	40.6	29.4	21.5
	OBC	39.5	37.8	30.0	20.7	17.1
	FC		23.0	17.6	12.5	10.5
	All	45.7	37.7	29.9	22.0	15.7

Source: Arvind Panagariya and Vishal More, 'Poverty by Social, Religious and Economic

Considering the State-wise poverty head count data from the NSS 61st and 68th rounds, it can be said that poverty is getting concentrated in few states and social groups. A group of four states comprising Chhattisgarh, Jharkhand, Arunachal Pradesh, Bihar, and Odisha had a highest proportion of poor in 2011-12. On the other hand, the states like Kerala, Himachal Pradesh, Punjab, Andhra Pradesh and Delhi shows the lowest rate of poverty.

The reduction in poverty rate between 2004-05 and 2011-12 is highest in Odisha (24.6 percentage points) followed by the states Uttarakhand (21.4 percentage point) and Maharashtra (21 percentage points). On the other hand, Assam, Jammu and Kashmir and Delhi registered the lowest reduction. The state Arunachal Pradesh registered an increase in rate of poverty from 31.4 percent in 2004-05 to 34.7 percent in 2011-12.

The poverty gap ratios have been incorporated in Table 21.5 separately for rural and urban areas. About 16.56 percent of the total consumption in the

rural areas in 1973-1974 was required to increase by 16.56 per cent to bring the poor to the poverty line whereas this came down to 9.63 percent in 2004-05 and further to 5.05 percent in 2011-12. The trend was the same in the urban areas where 13.64 percent of the total consumption was needed to bring the poor to the poverty line in 1973-74, 6.12 percent in 2004-05 and 2.7 percent in the year 2011-12 (Table 21.5).

**Table 21.5: Poverty Gap Ratio (Percentage)
(MRP Consumption Distribution)**

Year	Rural	Urban
1973-1974	16.56	13.64
1977-1978	15.73	13.13
1983	12.32	10.61
1987-1988	9.11	9.94
1993-1994	8.45	7.88
1999-2000	5.11	4.84
2004-2005	9.63	6.12
2011-12	5.05	2.7

Source: Estimated from the household consumer expenditure data of the NSSO, various Rounds.

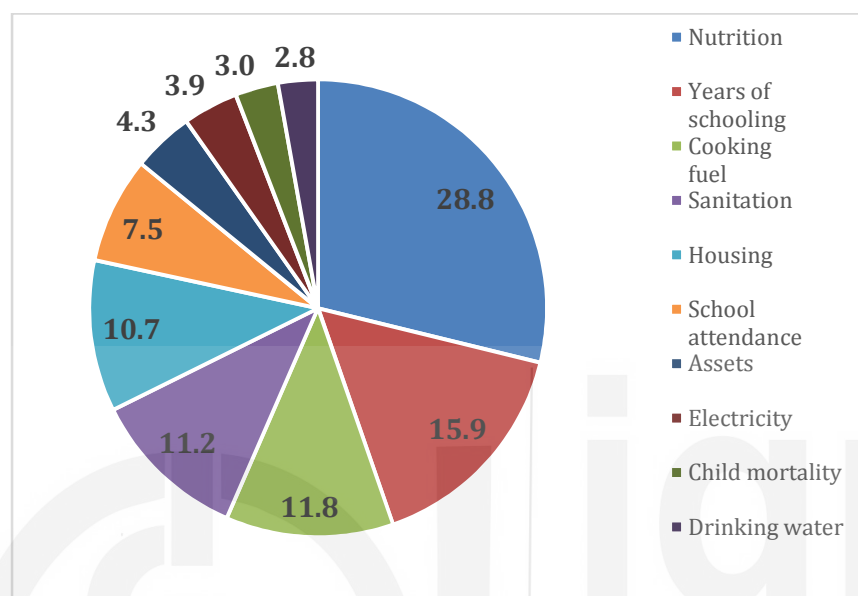
b) Income and Non-income Indicators of Poverty

So far, we have discussed poverty expressed by way of poverty ratio. No doubt it is an important aspect of the living standard, but this does not reflect certain aspects of non-income poverty expressed by way of the measures like health, education, nutrition deficiencies, human development outcomes, gender inequality status, etc. Studies show the headcount ratio as discussed in this section as a static measure and there are chances that people may fall back into poverty. Again, other studies highlighted education, health and nutrition are other dimensions that need to be taken into account. Let us discuss the Indian situation in terms of non-income measure of poverty.

The multidimensional poverty index as developed by Oxford Poverty and Human Development Initiative (OPHI) is a multidimensional poverty measure (MPI) which takes into account three dimensions i.e., health, education and living standard. Within health they have taken two indicators such as nutrition and child mortality, in education two indicators i.e., year of schooling and school attendance and in living standard six indicators such as cooking fuel, sanitation, drinking water, electricity, housing and asset. All the dimensions are given equal weight and within the dimension all the variables are equally weighted.

As per the finding of OPHI multidimension poverty indicators (using NFHS database) in India, poverty declined over the decade from 2005/6 to 2015/16 – from 635 million to 364 million poor persons. Again, it is interesting to find that out of 364 million MPI poor in India about 35 per cent are children. The

Multi-dimensional poverty indicators show that 27.9 percent of population are MPI poor. As compared to the income poor of 21.9 percent, the MPI poor shows a 6-percentage point higher poor. This method also calculates the contribution of 10 indicators in overall poverty. For the year 2015-16 the contribution of different dimensions to overall poverty is given in Figure 21.1.



Source: OPHI (2020)

Fig. 21.1: Indicator Contribution to Overall Poverty by Area

The figure clearly indicates that nutrition contributes 29 percent to the total poverty followed by the years of schooling (16 per cent) and cooking fuel and sanitation (each 11 per cent). The variables contributing least to overall poverty are drinking water and child mortality (each 3 per cent).

The state-level MPI poor indicate a large inequality in MPI poor among different states. The states like Bihar (52 per cent), Jharkhand (47 per cent), Madhya Pradesh (41 per cent), Uttar Pradesh (41 per cent) have the highest poverty head count in terms of MPI poor. On the other hand, the state registering the lowest proportion of MPI poor people are Kerala (1 per cent), Delhi (4 per cent), Punjab (6 per cent) and Tamil Nadu (7 per cent). In 2015-16, the four poorest states – Bihar, Jharkhand, Uttar Pradesh, and Madhya Pradesh – were still home to 196 million MPI poor people – over half of all the MPI poor people in India. Of the total MPI poor in India, 8.6 per cent still live in severe poverty.

The HDR published in the year 2019 considered four indicators in the index, viz. expectancy at birth, expected years of schooling, mean years of schooling, GNI per capita. As per the methodology if the HDI is approaching towards 1 the higher the status and if it approaching towards 0 the lower the progress of the country. In 2019 the Human Development index for India increased from 0.431 in 1990 to 0.647 in 2018 (Table 21.6)

Table 21.6: HDI Value in India 1990-2018

Year	HDI Value
1990	0.431
1995	0.463
2000	0.497
2005	0.539
2010	0.581
2015	0.627
2016	0.637
2017	0.643
2018	0.647

Source: UNDP HDR 2019 downloaded from
http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/IND.pdf

It is clear from the non-income poverty analysis that the nutrition constitutes an important dimension to the overall non-income poverty. In the next section, we will analyse malnutrition as a non-income poverty measure and its comparison to the income poverty measure.

Check Your Progress 1

- 1) How is the PG index more useful vis-à-vis poverty head count measure in assessing poverty?

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- 2) Identify the measures that take note of income as well as non-income aspects of poverty.

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- 3) What are the indicators used in capability poverty measure or a human development measure such as HDI?

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- 4) What do you mean by MPI? What are the advantages of MPI poverty measure over income poverty measure?

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21.5 THE CONCEPT OF MALNUTRITION

In this section, we shall analyse the concept of malnutrition and the status of India in terms of malnutrition measures. We discuss the type of malnutrition and different measures of malnutrition for both adult and children. First, we define the most common measures of nutrition to identify the nutrition outcome variables. Second, we discuss the direction of influence between different socio-economic indicators like wealth, social category, religious category, education on malnutrition both by reviewing relevant literature and the database. Third, we identify the main determinant variables from the literature and from other studies that are responsible for high malnutrition in India. These three aspects will be analysed and is compared with income poverty. Here the malnutrition indicators that we have used are malnutrition among children below 5 years comprising anthropometric measurement (stunting, wasting and underweight) and anaemia, whereas among adult we analyse prevalence of anaemia and Body Mass Index (BMI) status for women in the age group 15 to 49 years. The reason behind taking the malnutrition among children and women is because in developing countries like India, children and adults are vulnerable to malnutrition because of low dietary intakes, infectious diseases, lack of appropriate care and inequitable distribution of food within the household. Also, during early childhood under-nutrition of children can cause low cognitive outcomes and low productivity during adulthood. Also, children respond faster to changes in food intakes and food shortages. Hence, the anthropometric measures of children are considered to be a good proxy of nutrition situation of the country.

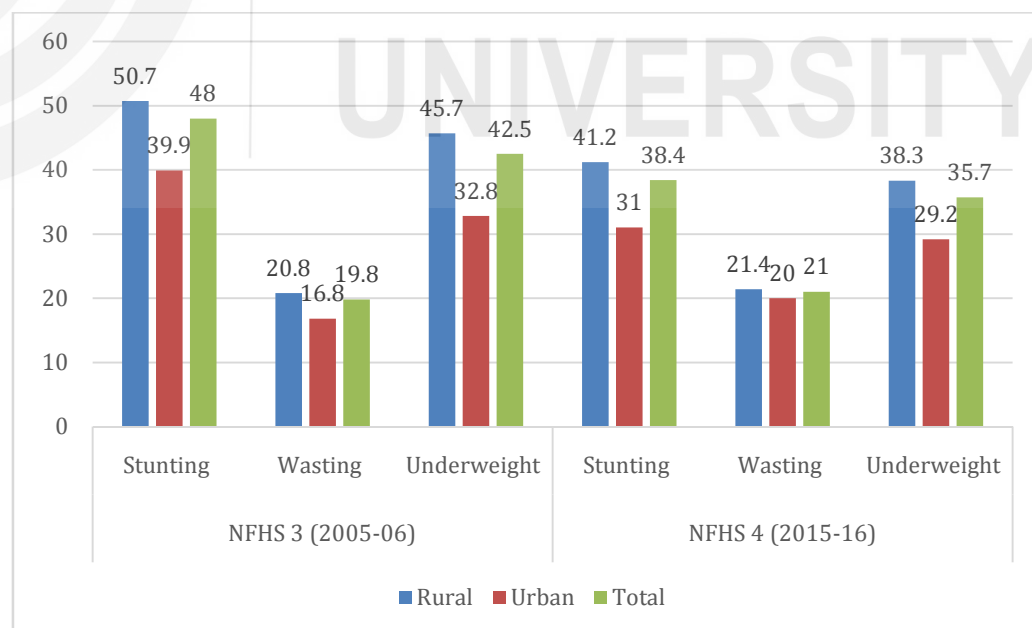
21.6 MALNUTRITION TYPE AND MEASUREMENT

This section is divided into two parts– the first part analysed the malnutrition among children and the second part analysed the malnutrition among women.

21.6.1 Malnutrition Type and Measurement among Children

In India, the percentage of stunted children in the age group of 0 to 5 years was 38.4 per cent in 2015-16 whereas in 2005-06 it was 48.0 percent which shows a decline by about 9.6 percent. The wasting in 2015-16 was 21 percent whereas in 2005-06 it was slightly lower at 19.8 percent. Again, the proportion of underweight which was 36 percent in 2015-16 declined by about 7 percent over one decade (2005-2015). Higher proportion of children was found malnourished in rural areas as compared to urban areas. In stunting and underweight the rural urban differential in the year 2015-16 was about 10 percentage point whereas in wasting it was only 1.5 percentage point difference.

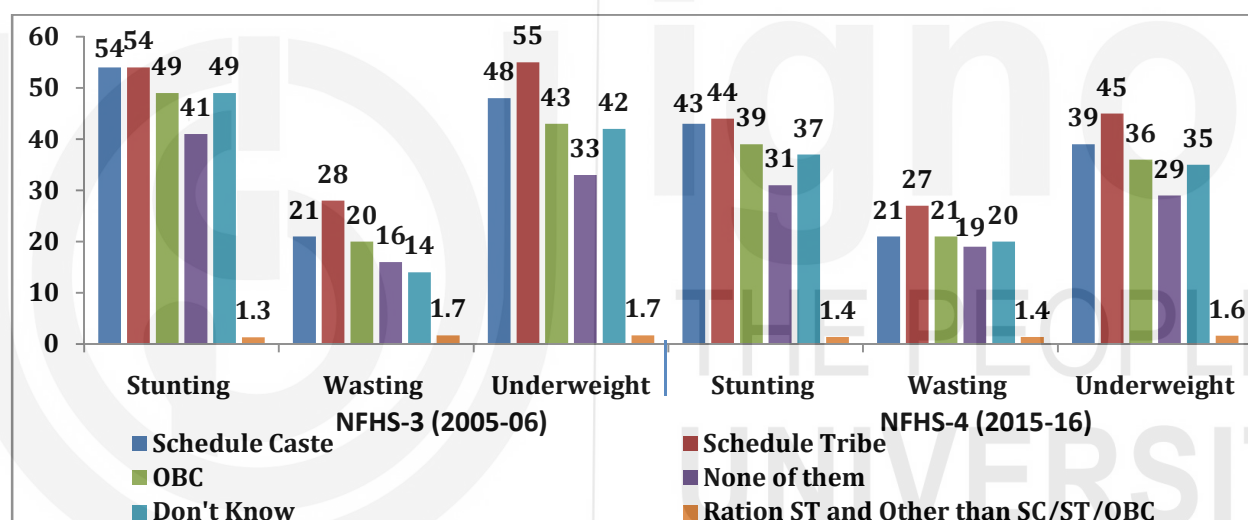
Assessing in terms of wealth index, in general, stunting among children declines, as a household moves up the income/wealth ladder (from poorest to richest). Over the last decade, percentage of stunted children has declined across every wealth quintile. In 2015-16, the proportion of stunted children belonging to the richest household declined to 22 percent from 26 percent in 2005-06. On the other hand, among poorest classes about half of the total children were stunted (52 percent) which decreased from 60 percent in 2005-06 (Figure 21.2).



Source: Calculated from NFHS 2005-06 and 2015-16.

Fig. 21.2: Trend Analysis of Malnutrition over a Decade (2005-06 to 2015-16)

As shown in Figure 21.3, in 2015-16, stunting among children has been found to be the highest among scheduled tribe households (44 percent) and lowest among the children belonged to other than SC/ST/OBC households. On the other hand, in the year 2005-06, a 13percentage point difference was found between children belong to SC category and other than SC/ST/ OBC category. In terms of wasting, in 2015-16 again ST children have the highest proportion of wasted children whereas it is lowest among children belonged to other than SC/ST/OBC category. The proportion of children underweight in 2015-16 is highest among STs (45per cent) and lowest among the children belonged to other than SC/ST/OBCs (29 per cent). Over the one-decade period the stunted among STs declined from 54 percent in 2005-06 to 44 percent in 2015-16. Whereas stunted children belonged to general caste declined from 41 percent in 2005-06 to 29 percent in 2015-16. In terms of underweight, in 2005-06the highest proportion was found among children belonged to STs (55 per cent) whereas it was lowest among general caste children (other than SC/ST/OBCs). The percentage point difference in underweight between ST and general children in 2005-06 was 21 percent whereas it was reduced to 16 percentage point change in 2015-16.



Source: Calculated from NFHS 2005-06 and 2015-16.

Fig. 21.3: Percentage of Children Stunted, Wasted and Underweight, by Social Category 2005-06 and 2015-16

Anaemia is a condition that is marked by low levels of haemoglobin in the blood. Iron is a key component of haemoglobin, and iron deficiency is estimated to be responsible for half of all anaemia globally (NFHS report 2015-16). Anaemia is a serious concern for children because it can impair cognitive development, stunt growth, and increase morbidity from infectious diseases.

As per NFHS data, the anaemia is a major concern and in 2015-16 the proportion of children anaemic is about 59 percent which is reduced by 11 percentage points as compared to 2005-06.

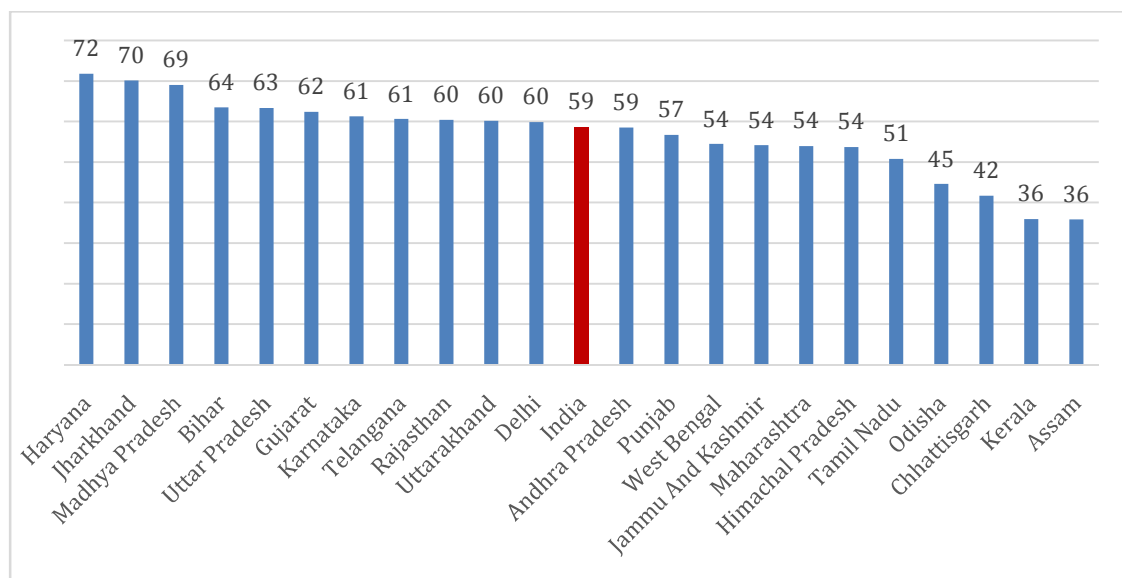
Table 21.7: Anaemia among Children 6-59 Months in India, 2005-06 and 2015-16

		NFHS-3	NFHS-4	NFHS 3-NFHS 4
Area	Urban	63.1	56.1	7.0
	Rural	71.6	59.6	12.0
Wealth	Poorest	76.6	64.1	12.5
	Poorer	73.7	59.9	13.8
	Middle	69.4	59.1	10.3
	Richer	64.7	54.5	10.2
	Richest	56.6	51.9	4.7
Social group	Schedule caste	72.5	60.8	11.7
	Schedule tribe	77.1	63.8	13.3
	OBC	70.2	58.7	11.5
	None of them	64.0	54.7	9.3
	Don't know	69.0	62.0	7.0
Religious group	Hindu	69.8	58.8	11.0
	Muslim	69.7	59.3	10.4
	Christian	60.4	45.8	14.6
	Sikh	63.9	56.6	7.3
	Buddhist/Neo-Buddhist	66.7	56.7	10.0
	Other	73.7	65.0	8.7
Total		69.5	58.7	10.8

Source: Calculated from NFHS 2005-06 and 2015-16

A large disparity was also visible. In 2015-16 the rural-urban differential reduced as compared to 2005-06. The anaemia among children by wealth quintile shows that among poorest the proportion was 64 percent as compared to the richest quintile of 52 percent. Again anaemia among children belonging to ST category is the highest among the social categories whereas the children belonged to the general caste population (other than SC/ST/OBCs) is the lowest. Among religious category, the highest proportion of anaemic children were Hindu and Muslims.

An analysis of anaemia among children 6-59 months across the major states of India shows that in 2015-16, anaemia was higher in the state of Haryana (72 per cent). This is peculiar as the state has low poverty incidence in terms of income poverty. The other states registering higher proportion of anaemic children are Jharkhand, Madhya Pradesh, Bihar and Uttar Pradesh. On the other hands in states of Assam (36per cent), Kerala (36per cent), Chattisgarh and Odisha it is the lowest.



Source: Calculated from NFHS 2005-06 and 2015-16

Fig. 21.4: Percentage of Children Anaemic by States, 2015-16

21.6.2 Malnutrition Type and Measurement among Women

As discussed earlier, BMI is one of the indicators to measure the nutrition status of women. As per the WHO standard, the BMI is calculated for women in the age group of 15-49 years. The proportion of thin women declined from 36 percent in 2005-06 to 23 percent in 2015-16 and at the same time the proportion of overweight or obese women increased from 13 percent to 21 percent. Overall, there has been an increase in the mean BMI from 20.5 in 2005-06 to 21.9 in 2015-16.

The proportion of thin women in rural and urban India are respectively 15.5 per cent and 26.8 per cent showing a 11 percentage point difference. As expected, the thin women by social category shows that STs and SCs bear the highest percentage (ST 31.7 per cent and SC 25.3 per cent) and it is lowest among general caste population (other than SC/ST/ OBC).

The other important indicator measuring the nutritional status of women is the anaemia. As per the NFHS statistics, the proportion of women anaemic was 53 percent in 2015-16 as compared to 55.3 percent in 2005-06. The anaemia among women by wealth quintile shows a variation with about 59 percent of women in the lowest wealth quintile are anaemic as compared to 48.2 percent in the highest wealth quintile. The anaemia of women by schooling status shows that among illiterate women, the proportion of women anaemic was 56.4 percent as compared to 48.7 percent for women having education at intermediate level.

21.7 MALNUTRITION AND POVERTY: A COMPARATIVE ANALYSIS

Hunger, malnutrition, hidden hunger and poverty all are related to each other. Poverty is considered to be the main cause of malnutrition and hunger,

however rich people are found to be malnourished as well. In this section we analyse the interaction between income poverty and malnutrition.

Table 21.8: A Comparative Analysis of Income Poverty and Non-Income Poverty by State

	Income Poverty 2011-12	Per capita NSDP 2017-18 at constant price	MPI Poor 2015-16	Anaemia Children 2015-16	Stunted Children 2015-16	Thin Women 2015-16
Kerala	7.1	138368	1	36.0	20	9.7
Himachal Pradesh	8.1	130644	8	53.7	26	16.2
Punjab	8.3	110802	6	56.7	26	11.7
Andhra Pradesh	9.2	103214	16	58.5	31	17.6
Delhi	9.9	255431	4	59.9	32	14.8
Jammu & Kashmir	10.4	62984	15	54.2	27	12.1
Haryana	11.2	159892	11	71.8	34	15.8
Uttarakhand	11.3	147204	17	60.2	34	18.4
Tamil Nadu	11.3	133029	7	50.8	27	14.6
Rajasthan	14.7	74441	32	60.4	39	27
Gujarat	16.6	142068	22	62.4	39	27.2
Maharashtra	17.4	140724	17	54.0	34	23.5
West Bengal	20.0	64007	26	54.5	33	21.3
Karnataka	20.9	143827	17	61.3	36	20.8
Uttar Pradesh	29.4	42798	41	63.3	46	25.3
Madhya Pradesh	31.7	54264	41	69.0	42	28.4
Assam	32.0	57835	36	35.9	36	25.7
Odisha	32.6	72760	36	44.6	34	26.5
Bihar	33.7	26699	52	63.5	48	30.5
Jharkhand	37.0	52277	47	70.1	45	31.6
Chhattisgarh	39.9	66122	37	41.7	38	26.7
All-India	21.9	87828	28	56.9	38	22.9

Table 21.8 presents the income and non-income poverty indicators. When we compare the income poverty and anaemia among children it shows that Andhra Pradesh and Delhi registered low-income poverty and high anaemia among children. On the other hand, Chhattisgarh, Odisha and Assam registered high proportion of income poverty but comparatively low proportion of anaemic children. Again, when we compare the income poverty and stunted children, Andhra Pradesh and Delhi which have lower poverty rate registered high proportion of stunted children. In terms of comparison of